Jacob R. Nesslage

Postdoctoral Scholar, eDNA Explorer Inc. · Visiting Scientist, University of California, Santa Cruz

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EDUCATION

Ph.D., Environmental Systems 2026
University of California, Merced · Emphasis: Ecology and Ecosystem Science
Dissertation: Biodiversity at the Terrestrial-Aquatic Interface: Quantification Through Integrated Earth Observation and Multi-Modal Biodiversity Surveys
Advisor: Dr. Erin L. Hestir · *Committee:* Dr. Joshua H. Viers, Dr. Rachel S. Meyer, Dr. Teamrat A. Ghezzehei

B.S., Environmental Engineering, *magna cum laude* 2021
University of California, Merced

A.S., Physics 2018
Modesto Junior College

APPOINTMENTS

Postdoctoral Scholar 2026–present
eDNA Explorer Inc.

Visiting Scientist 2026–present
University of California, Santa Cruz

RESEARCH EXPERIENCE

Graduate Student Researcher 2021–2026
University of California, Merced · Advisor: Dr. Erin L. Hestir

- Collected RGB, multispectral, and hyperspectral data in terrestrial and aquatic environments using field spectrometers and UAVs
- Conducted field surveys and analysis of eDNA, soil, water, and plant samples
- Built GIS workflows in ArcGIS and QGIS and developed the OpenRES QGIS plugin
- Developed R, Python, JavaScript, and Bash tools to process large remote sensing datasets, perform statistical modeling, and develop sensor-fusion algorithms

Undergraduate Remote Sensing and GIS Research Assistant 2020–2021
University of California, Merced · Advisor: Dr. Erin L. Hestir

- Assisted with UAV collection of hyperspectral imagery and LiDAR, field spectroradiometry, and water sampling
- Developed R and Python scripts for remote sensing data processing

Natural Resources Specialist 2019
USDA Natural Resources Conservation Service · Supervisor: Ted Strauss, State Air Quality Conservationist

- Supported voluntary incentive programs helping landowners reduce emissions of ozone precursors, particulate matter, and greenhouse gases
- Quantified program outcomes under the State Implementation Plan and reported results to CalEPA

Instructional Laboratory Assistant 2019
University of California, Merced · Supervisor: Dr. Luke Reed

- Supported engineering capstone and graduate projects with 3D printing, laser cutting, CNC, and injection molding
- Maintained laboratory experiments in circuits, fluid mechanics, materials, sensors and actuators, and mechatronics

PEER-REVIEWED PUBLICATIONS

1. **Nesslage, J.R.**, Hestir, E.L., and Viers, J.H. (2026). Plant species richness responses to microtopography across a vernal pool complex revealed by UAV-LiDAR. *Environmental Research: Ecology*, 5(1).
2. Hayden, M.T., Rossi, M.W., Dee, L.E., Kovach, K., Amaral, C.H., **Nesslage, J.**, Slimp, M., Meyer, R.S., and Stavros, E.N. (2026). Scale dependence in remotely sensed biodiversity: Leveraging continental-scale imaging spectroscopy from the National Ecological Observatory Network. *Remote Sensing in Ecology and Conservation*.
3. Cardoso, A.W., Hestir, E.L., Slingsby, J.A., Forbes, C.J., Moncrieff, G.R., Turner, W., Skowno, A.L., **Nesslage, J.**, Brodrick, P.G., Gaddis, K.D., and Wilson, A.M. (2025). The biodiversity survey of the Cape (BioSCape), integrating remote sensing with biodiversity science. *npj Biodiversity*, 4(1), 2.
4. Cardoso, A.W., Hestir, E.L., Slingsby, J.A., Forbes, C.J., Moncrieff, G.R., Turner, W., Skowno, A.L., **Nesslage, J.**, Brodrick, P.G., Gaddis, K.D., and Wilson, A.M. (2025). BioSCape combines local knowledge and remote-sensing technology for inclusive biodiversity science. *Nature Reviews Biodiversity*, 1(1), 2–4.

PEER-REVIEWED CONFERENCE PAPERS

1. **Nesslage, J.**, Barreto, B.L., Weingram, A., and Hestir, E. (2023). A machine learning approach for high resolution fractional vegetation cover estimation using Planet CubeSat and RGB drone data fusion. In *IGARSS 2023 – IEEE International Geoscience and Remote Sensing Symposium* (pp. 4879–4882). IEEE.
2. Cardoso, A., Hestir, E., Slingsby, J., **Nesslage, J.**, Forbes, C., and Wilson, A. (2023). BioSCape: Combining airborne hyperspectral and laser altimeter with field data to advance remote sensing of biodiversity. In *IGARSS 2023 – IEEE International Geoscience and Remote Sensing Symposium* (pp. 499–502). IEEE.

SUBMITTED AND IN PREPARATION

1. **Nesslage, J.R.** and Hestir, E.L. Open Riverine Ecosystem Synthesis (OpenRES): A QGIS plugin for automated extraction of hydrogeomorphic features to support Functional Process Zone classification of river networks. *Submitted to Journal of Open Source Software.*
2. Li, L., Wang, Z., **Nesslage, J.R.**, and Zhang, B. Mitigating bias in biodiversity patterns using deep learning. *Submitted to Ecological Monographs.*
3. **Nesslage, J.R.**, Quick-Cleveland, J., Meyer, R.S., Viers, J.H., and Hestir, E.L. Soil-based eDNA metabarcoding and remote sensing elucidates baseline diversity and habitat preferences of multiple ecological communities across a multi-benefit floodplain restoration site. *In preparation.*
4. **Nesslage, J.R.**, Meyer, R.S., Worth, A., Newcomer, M., and Hestir, E.L. Functional Process Zones quantify hydrogeomorphic organization of Mediterranean-montane river benthic macroinvertebrate communities: Comparison across conventional and molecular survey. *In preparation.*

SOFTWARE

1. **Nesslage, J.** (2026). ShinyFPZ: An R Shiny app to aid in Functional Process Zone classification (v1.0.0). Zenodo. License: MIT.
2. **Nesslage, J.** (2026). STFuse: A Python package for spatiotemporal fusion of remote sensing imagery. Private repository (pending invention disclosure, UC Case No. C-2026-619-1).
3. **Nesslage, J.** and Hestir, E. (2025). OpenRES (Open Riverine Ecosystem Synthesis): A QGIS plugin for automated extraction of hydrogeomorphic features to support functional process zone classification of river networks. Zenodo. License: GPL-3.0.

GRANTS, FELLOWSHIPS, AND AWARDS

Group Achievement Award, BioSCape Airborne Remote Sensing Observation Team 2024–2025
National Aeronautics and Space Administration (NASA)

University of California President's Pre-Professoriate Fellowship (UC-PPFP) 2024–2025
University of California Office of the President · $37,000 + $10,000 stipend

National Science Foundation Graduate Research Fellowship (NSF-GRFP) 2021–2024
U.S. National Science Foundation · $132,000

Environmental Systems Summer Fellowship 2025
University of California, Merced · $8,500

Environmental Systems Travel Award 2022
University of California, Merced · $1,000

Calvin E. Bright Engineering Scholarship 2021
University of California, Merced · $3,000

UC Merced Renewable Energy Scholarship 2019, 2020
University of California, Merced · $1,500

John Elia Veterans Scholarship 2019, 2020
University of California, Merced · $2,000

Francis E. Benton Scholarship 2019, 2020
University of California, Merced · $1,000

Peter & Rochelle Koch Scholarship 2018
Valley Land Alliance · $500

CONFERENCE PRESENTATIONS

1. **Nesslage, J.**, Adepoju, S., and Hestir, E. (2026). Spatiotemporal fusion of drone and satellite imagery: Validation and optimization for row and tree crops. Internet of Things for Precision Agriculture Annual Meeting, Purdue University, West Lafayette, IN. *(Flash talk)*
2. **Nesslage, J.**, Adepoju, S., Dilmurat, K., and Hestir, E. (2025). Spatiotemporal fusion of commercial SmallSat and drone imagery using spectral unmixing and weighting-based algorithms. Internet of Things for Precision Agriculture NSF Site Visit, Pennsylvania State University, Philadelphia, PA. *(Poster)*
3. **Nesslage, J.**, Devarasetty, A., Adepoju, S., Dilmurat, K., and Hestir, E. (2025). Parameterization of unmixing and weighting-based spatiotemporal fusion methods using PlanetScope and UAV imagery. Internet of Things for Precision Agriculture Annual Meeting, Pennsylvania State University, Philadelphia, PA. *(Flash talk)*
4. **Nesslage, J.**, Hestir, E.L., Rossi, M.W., Meyer, R., and Stavros, E.N. (2024). Functional Process Zone delineation for riverine biodiversity assessment: QGIS plugin development and case studies across two continents. AGU24, Washington, D.C. *(Oral)*
5. **Nesslage, J.**, Quick-Cleveland, J., Meyer, R.S., Hestir, E.L., and Viers, J.H. (2024). Riparian patches as refugia within floodplain agroecosystems: Insights from two Central Valley case studies. Labor and Automation in California Agriculture Annual Meeting, University of California, Davis. *(Poster)*
6. **Nesslage, J.** and Hestir, E.L. (2024). Estimating plant species richness in an ephemeral wetland using UAV-LiDAR derived hydrologic and topographic data and machine learning. Ecological Society of America Annual Meeting, Long Beach, CA. *(Poster)*
7. **Nesslage, J.** (2024). Estimating vernal pool plant diversity using remote sensing. Merced Vernal Pools and Grassland Reserve 10th Anniversary & Science Symposium, Merced, CA. *(Invited talk)*
8. **Nesslage, J.**, Quick-Cleveland, J., Meyer, R.S., Hestir, E.L., and Viers, J.H. (2023). Monitoring Mediterranean floodplain restoration using multi-sensor fusion and environmental DNA. NASA Carbon Cycle and Ecosystems Joint Science Workshop, University of Maryland, College Park, MD. *(Poster)*
9. **Nesslage, J.**, Hestir, E., Quick-Cleveland, J., and Meyer, R. (2022). Monitoring taxonomic and functional (agro)biodiversity along the agricultural-riparian matrix of California's San Joaquin River using remote sensing and environmental DNA. AGU Fall Meeting, Chicago, IL (B22D-1468). *(Poster)*
10. **Nesslage, J.**, Quick-Cleveland, J., Meyer, R.S., Hestir, E.L., and Viers, J.H. (2022). Using eDNA and remote sensing to quantify species diversity, functional diversity, and their environmental drivers along an agricultural-riparian gradient. Ecological Society of America Annual Meeting, Montreal, QC. *(Poster)*
11. **Nesslage, J.**, Quick-Cleveland, J., Meyer, R.S., Hestir, E.L., and Viers, J.H. (2022). Functional agrobiodiversity monitoring using environmental DNA and remote sensing data fusion. FIRA 2022, Fresno, CA. *(Poster)*

12. **Nesslage, J.**, Fortier, M., and Hestir, E. (2022). A life cycle assessment of UAV remote sensing platforms. Internet of Things for Precision Agriculture Annual Meeting, Purdue University, West Lafayette, IN. *(Poster)*
13. **Nesslage, J.**, Arizpe Arroyo, D., and Hestir, E. (2022). Sensor fusion for multiscale applications: Case studies and future directions. Internet of Things for Precision Agriculture Annual Meeting, Purdue University, West Lafayette, IN. *(Poster; third place, poster competition)*
14. **Nesslage, J.**, Hestir, E., and Burmistrova, J. (2020). Using Sentinel-2 to detect and predict biodiversity hotspots in a vernal pools and grasslands ecosystem. American Geophysical Union Annual Symposium, San Francisco, CA. *(Poster)*

TEACHING EXPERIENCE

Guest Lecturer, Proximal and Remote Sensing Technologies for Soil and Plant Sciences Fall 2025
Fresno State University (Instructor of Record: Dr. Sharon Benes) · Remote sensing of biodiversity in vernal pools and floodplains

Guest Lecturer, ENGR 180: Spatial Analysis and Modeling Fall 2025
University of California, Merced (Instructor of Record: Dr. Erin Hestir) · Remote sensing of biodiversity in vernal pools and floodplains

Teaching Assistant, ENGR 180: Spatial Analysis and Modeling Spring 2025
University of California, Merced (Instructor of Record: Dr. Crystal Kolden) · GIS fundamentals, spatial modeling, and ArcGIS proficiency

Guest Lecturer, ENGR 180: Spatial Analysis and Modeling Fall 2024
University of California, Merced (Instructor of Record: Dr. Erin Hestir) · Remote sensing of biodiversity

Course Co-Creator, ES 292: Introduction to Bayesian Statistics for Environmental Systems Spring 2024
University of California, Merced (Instructor of Record: Dr. Erin Hestir) · Bayesian statistics and modeling using R and Stan

Instructor, Generalized Dissimilarity Modeling Workshop Summer 2023
University of California, Merced · One-week intensive workshop on generalized dissimilarity modeling in R

Guest Lecturer, ENVE 152: Remote Sensing of the Environment Fall 2022
University of California, Merced (Instructor of Record: Dr. Erin Hestir) · Remote sensing of biodiversity

Guest Lecturer, Bobcat Summer Academy Summer 2022
University of California, Merced · Using remote sensing to study environmental problems, for a 5th-grade audience

Guest Lecturer, ME 190: Novel Technologies in Agriculture Fall 2021
University of California, Merced (Instructor of Record: Dr. Reza Ehsani) · UAV remote sensing for agricultural problems

CERTIFICATIONS

Engineer-in-Training (EIT), CA Environmental Engineering, No. 187843 2026
California Board for Professional Engineers, Land Surveyors, and Geologists

NASA Open Science 101 2025
National Aeronautics and Space Administration

Motorboat Operator 2021
Scientific Boating Safety Association, UC Davis Bodega Marine Lab

Part 107 sUAS Remote Pilot 2021
Federal Aviation Administration

SKILLS

- **GIS:** QGIS, ArcGIS, GRASS GIS
- **Remote sensing image analysis:** ENVI, SNAP, ACOLITE, UgCS, Pix4D, OpenDroneMap
- **Programming:** Proficient in R, Python, and Unix shell; literate in JavaScript

- **Drone/UAV operations:** DJI Phantom 4 Pro, DJI Matrice 300, DJI Matrice 600
- **Surveying equipment:** Topcon and Trimble GPS systems; Topcon terrestrial LiDAR

PROFESSIONAL MEMBERSHIPS

- IEEE Geoscience and Remote Sensing Society
- American Geophysical Union
- Ecological Society of America

MILITARY SERVICE

Healthcare Specialist / Medic (68W) 2011–2014

3d Cavalry Regiment, U.S. Army · Fort Hood, TX

- Provided medical care in acute and non-acute situations to military personnel in clinical and field settings
- Taught Combat Lifesaver courses to deploying personnel
- Led a three-person team in evacuation operations, providing first-responder support for training exercises and range operations